



Session 2425W (June – July 2026)

PROJECT REPORT

Analysis of Player and Team Performance Trends in the IPL (2008–2026)

An Interactive Power BI Dashboard

Academic Task	2 (Project) — Nature: Individual
Course Title	Data Analytics Essentials with AI: SQL, Python, Power BI, Tableau and Excel
Course Code	PETV103
Section	9PV24
Maximum Marks	30
Date of Allotment	June 29, 2026
Date of Submission	July 16, 2026
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1. Introduction

The Indian Premier League (IPL) is the most widely watched T20 cricket league in the world. Since its inception in 2008, the IPL has generated an enormous volume of structured data – covering matches, teams, players, and ball-by-ball events – across nineteen seasons. Every match produces information on toss decisions, venues, results, and player performances, while every delivery bowled adds further granularity in the form of runs, boundaries, and wickets.

Proper analysis of this data allows fans, analysts, franchises, and broadcasters to understand team performance, identify consistently high-performing players, evaluate the impact of toss decisions, and track how the game itself is evolving over time — for example, the rising trend of boundary-hitting in recent seasons. Data analytics converts these raw records into structured insights, while visualization communicates these insights quickly and intuitively through charts, KPIs, and interactive filters, rather than requiring the reader to interpret raw tabular data.

This project was undertaken to explore IPL data because cricket combines rich statistical depth with wide public interest, making it an ideal subject to practice the complete data analytics workflow – from data cleaning and modelling to dashboard design and storytelling. This project develops an interactive two-page Power BI dashboard that converts raw IPL match and delivery data into a clear, explorable story of team success, toss trends, venue patterns, and the league's evolving scoring style.

2. Objectives

- To analyse match-level trends across all IPL seasons from 2008 to 2026.
- To identify the most successful and most consistent teams by total wins and win percentage.
- To study the impact of the toss — and the toss decision (bat/field first) — on match outcomes.
- To identify the top Player of the Match performers across IPL history.
- To analyse venue-wise and city-wise distribution of matches.
- To examine scoring trends, including total runs, fours and sixes, across seasons.
- To design and build an interactive, story-driven Power BI dashboard with KPIs, charts, and slicers.
- To generate meaningful, data-backed insights and recommendations from the analysis.

3. Problem Statement

Although IPL data is publicly available in large volumes, it is scattered across multiple raw files (match records, ball-by-ball deliveries, team and player master data) and is not easy for a fan, analyst, or franchise stakeholder to interpret at a glance. There is no single, interactive view that brings together team performance, toss impact, venue trends, and scoring evolution in one place. The objective of this project is to consolidate these datasets into a connected data model and build an interactive Power BI dashboard that supports quick, story-driven decision-making – enabling a user to filter by season, team, venue, or player and instantly see how outcomes change.

4. Dataset Description

Item	Description
Dataset Name	IPL Seasons 2008 to 2026 Dataset
Source	Kaggle
Number of Tables	6 linked tables (see Section 5)
Core Fact Table	ipl_matches_data — 1,212 rows $\times$ 26 columns
Ball-by-Ball Table	ball_by_ball_data — 288,226 rows $\times$ 30 columns
Players Table	players-data-updated — 799 rows $\times$ 6 columns
Teams Table	teams_data — 16 rows $\times$ 2 columns
Time Period	2008 to 2026 (19 seasons; the 2026 season is still in progress at the time of this report, with 43 matches recorded so far)

Important variables used in the analysis:

- match_id, season, match_date — unique identifiers and timing of each match
- team1, team2, toss_winner, toss_decision, match_winner — match participants and outcome
- venue, city — location of each match
- result, win_by_runs, win_by_wickets — margin and nature of victory
- player_of_match — award winner for each match

- batter, bowler, batter_runs, total_runs, is_wicket, over_number — ball-by-ball scoring detail

5. Data Source

Website	Kaggle
Dataset URL	https://www.kaggle.com/datasets/slidescope/ipl-seasons-2008-to-2025-dataset
Downloaded On	July 5, 2026
License	Apache 2.0
Files Used	ipl_matches_data.csv, ball_by_ball_data.csv, players-data-updated.csv, teams_data.csv, team_aliases.csv, IPL_finals.csv

6. Data Cleaning and Preparation

The six source files were imported into Power BI through Power Query and connected using the following pre-processing steps:

- Loaded and typed all six CSV files (ipl_matches_data, ball_by_ball_data, players-data-updated, teams_data, team_aliases, IPL_finals), explicitly setting correct data types for each column (integer IDs, text, date, and logical/boolean fields such as is_wicket and is_wide_ball).
- Converted match_date to a proper Date type and enabled a Date table for time-based analysis.
- Built a separate team_aliases table to map historically renamed franchises (e.g., Delhi Daredevils → Delhi Capitals, Kings XI Punjab → Punjab Kings) back to a single team, so historical performance can be tracked under the current team identity.
- Created relationships linking team1, team2, toss_winner, match_winner, and player_of_match in ipl_matches_data to their respective lookup tables (teams_data, players-data-updated) so that IDs display as readable names in visuals.
- Added calculated columns: LosingTeam (identifies the losing team ID by comparing match_winner to team1/team2), “Losing TeamName” (resolves this to a team name), and IsFinal (flags whether a match was a season final, using the IPL_finals table).

- Created DAX measures for key metrics: “4s” and “6s” (counts of boundary deliveries from ball_by_ball_data), “Num of Balls” (total delivery count), and “Num of Overs” (total balls ÷ 6).
- Reviewed the data for missing values — win_by_runs and win_by_wickets are blank for tied or no-result matches, which is expected and was left as-is rather than treated as an error.

7. Methodology

The project followed a structured analytics workflow:

- Data Collection — six related CSV files downloaded from Kaggle.
- Data Cleaning — performed in Power Query (type correction, relationship keys, alias mapping).
- Data Modelling — star-schema style model connecting the match fact table to team, player, alias, finals, and date dimension tables.
- DAX Measure Creation — boundary counts, ball/over totals, and supporting calculated columns.
- Dashboard Design — a two-page interactive Power BI report combining KPI cards, charts, a table, and slicers.
- Exploratory Analysis — interpreting each visual to draw out trends and patterns.
- Insights and Conclusion — translating the findings into recommendations.

Software used: Power BI Desktop (Power Query for data preparation, the Tabular data model for relationships, and DAX for measures and calculated columns).

8. Dashboard Design

Live Dashboard Link: <https://app.powerbi.com/view?r=eyJrIjoiaYzliY2Y2OTctMDNkOC03D%3D>

GitHub Repository: <https://github.com/kumaradi9508/IPL-Performance-Analytics->

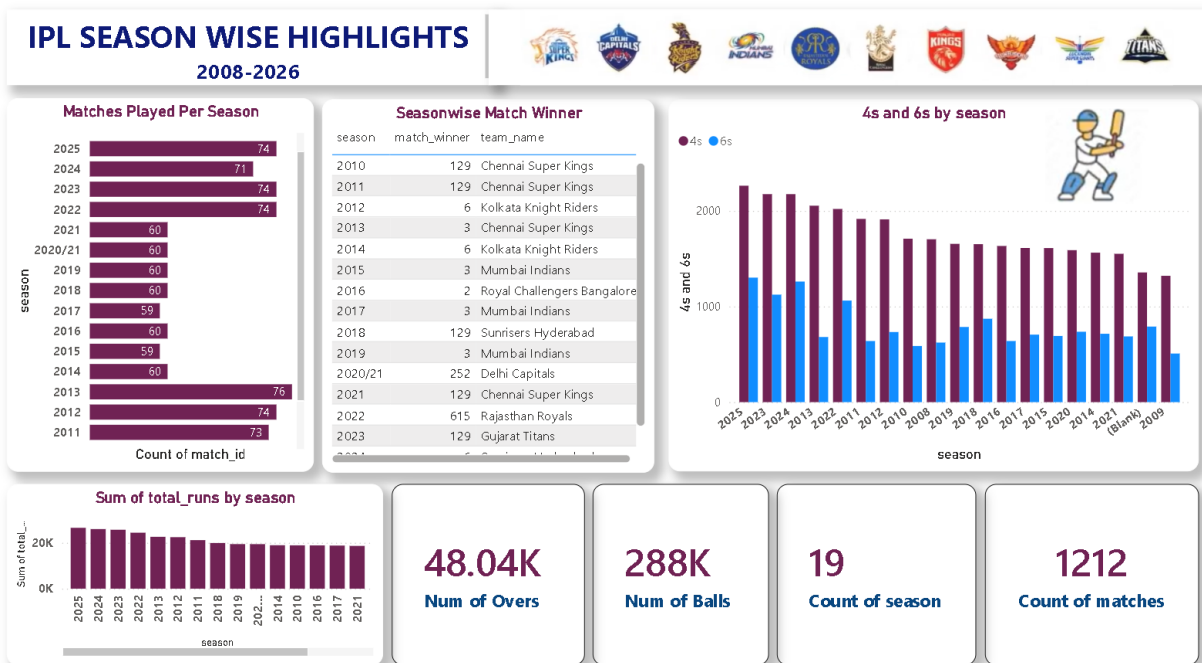


Figure 1: IPL Season Wise Analysis 2008-2026

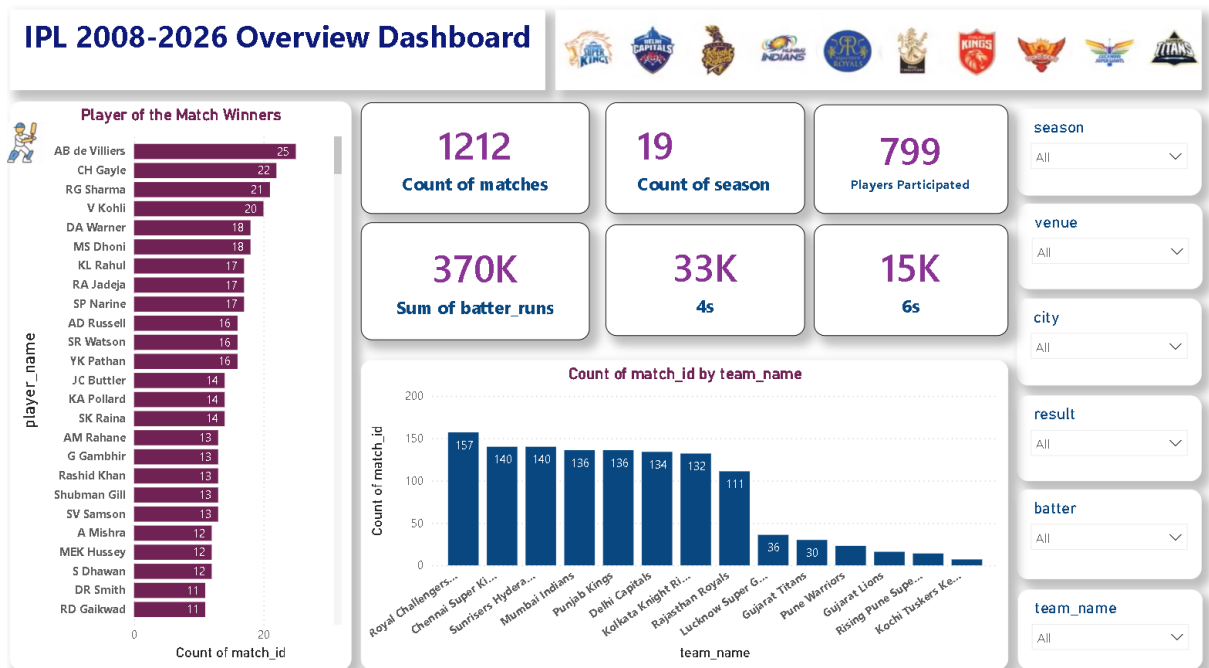


Figure 2: IPL 2008-2026 Overview Dashboard

Page 1 — “IPL Season Wise Highlights 2008–2026”

KPI Cards: Total Matches · Total Seasons · Total Balls Bowled (Num of Balls) · Total Overs (Num of Overs)

Charts: Stacked Bar chart of Matches Played per Season · Table of Season-wise Match Winners · Clustered column chart of fours and sixes hit in every season · Clustered column chart of total runs scored per season

Supporting visuals: Team logo images placed for visual identity.

Page 2 — “IPL 2008–2026 Overview Dashboard”

KPI Cards: Total Matches · Total Season · Total Players Participated · Total Batter Runs · Total Fours · Total Sixes

Charts: Bar chart of Player of the Match winners · Clustered column chart of matches by team

Interactive Slicers: Season, Venue, City, Match Result, Batter, and Team — allowing any visual on the page to be filtered by these dimensions, so a viewer can, for example, select a single season and city to instantly see that slice of the story.

9. Analysis and Findings

Matches and Seasons

- The dataset covers 1,212 matches across 19 seasons (2008–2026), played across 59 venues in 38 cities.
- Matches per season rose from 58 in 2008 to a peak of 76 in 2013, then stabilised around 60 per season through the mid-2010s, before rising again to 71–74 from 2022 onward following the league’s expansion to 10 teams.
- The 2026 season is still in progress, with only 43 matches recorded at the time of this analysis — it should not be compared directly with completed seasons.

Team Performance

- By total wins, Mumbai Indians lead with 155 wins, followed by Chennai Super Kings (146), Royal Challengers Bangalore (139), Kolkata Knight Riders (137), Sunrisers Hyderabad (132), Punjab Kings (130), Delhi Capitals (126), and Rajasthan Royals (125).
- By win percentage, however, Gujarat Titans lead the league at 60.9%, ahead of Chennai Super Kings (56.2%) and Mumbai Indians (54.4%) — showing that a newer franchise can outperform historical totals on a per-season consistency basis.

Toss Analysis

- Teams chose to field first in 800 of 1,212 matches (about 66%), compared to batting first in only 412 matches (34%) — a strong, consistent preference toward chasing, likely driven by dew and better visibility of the target.
- The team that won the toss also won the match in 51.7% of games — only marginally better than a coin flip, indicating the toss gives a slight but not decisive advantage.

Match Results

- Of all matches, 1,187 produced a clear winner, while only 16 ended in a tie (1.3%) and 9 had no result, typically due to weather.

Player Performance

- AB de Villiers has won the most Player of the Match awards (25), followed by Chris Gayle (22), Rohit Sharma (21), and Virat Kohli (20) — with David Warner and MS Dhoni tied next at 18 each, and KL Rahul, RA Jadeja and Sunil Narine tied at 17 each — reflecting sustained individual impact across many seasons.

Venue Analysis

- Eden Gardens has hosted the most matches (77), followed by the Wankhede Stadium (73 under one naming variant, plus 57 under a second spelling — see Limitations), M. Chinnaswamy Stadium (65), and Feroz Shah Kotla (60) — confirming that traditional home venues have hosted the bulk of IPL cricket.

Scoring Trends

- Across all recorded deliveries, batters have scored a combined 370,266 runs from 288,226 balls bowled, with 33,469 fours and 15,143 sixes struck league-wide.
- Run-scoring has increased sharply in recent seasons: total runs per season rose from roughly 18,000–22,000 in the 2008–2018 period to 24,000–26,500 per season from 2022 onward.
- Sixes per season nearly doubled from 687 in 2021 to 1,302 in 2025, and fours rose from 1,551 to 2,263 over the same period — reflecting more aggressive batting, likely influenced by the Impact Player rule introduced in 2023.

10. Business Insights / Recommendations

- Franchises evaluating team-building strategy could study Gujarat Titans' model, since its win percentage (60.9%) outperforms teams with far longer histories.
- Captains should continue to strongly favour fielding first at the toss, given its 66% historical preference, while recognising that the toss itself only provides a marginal (51.7%) edge — match strategy matters more than the toss outcome.
- Broadcasters and sponsors can spotlight consistently high-impact players such as Man-deep Singh and N. Rana for promotional and brand campaigns, since award consistency signals reliable fan-facing performance.
- High-capacity, traditional venues (Eden Gardens, Wankhede Stadium, M. Chinnaswamy Stadium) should continue to be prioritised for marquee and playoff fixtures, given their proven hosting history.
- The data team should standardise venue names (e.g., merging “Wankhede Stadium” and “Wankhede Stadium, Mumbai”) and team-name spelling variants (e.g., “Rising Pune Supergiant” vs. “Rising Pune Supergiants”) to avoid double-counting in any future analysis.
- Given the sharp rise in fours and sixes since 2022, coaching and scouting decisions should place growing emphasis on power-hitting skills, consistent with the league's evolving, higher-scoring format.

11. Conclusion

This project demonstrates the complete data analytics workflow — from consolidating six related IPL datasets, cleaning and modelling them in Power BI, to building a two-page interactive dashboard that tells a clear analytical story. The analysis shows that while Mumbai Indians and Chennai Super Kings remain the most successful teams by total wins, newer franchises like Gujarat Titans can be more consistent on a win-percentage basis. Toss decisions strongly favour fielding first, though the toss itself is only a marginal advantage. Most notably, the league's scoring style has changed considerably in recent seasons, with boundary-hitting rising sharply since 2022. Interactive dashboards such as this one allow fans, analysts, and franchises to explore these patterns directly rather than relying on static reports, supporting faster and more informed interpretation of the game.

12. Limitations

- The 2026 season is incomplete (only 43 matches recorded), so any season-wise comparison involving 2026 understates its true totals.
- Minor data inconsistencies remain in the raw source — for example, duplicate venue name spellings and a duplicated team name (“Rising Pune Supergiant” vs. “Rising Pune Supergiants”) — which were not fully standardised through the team_aliases table.
- The dataset does not include player auction price or salary data, so the financial/ROI dimension of team performance is outside the scope of this analysis.
- External factors such as injuries, pitch conditions, and weather (beyond “no result” flags) are not captured in the data and could not be analysed.

13. Future Scope

- Fully standardise team and venue names using the team_aliases table so that year-over-year comparisons are not affected by naming inconsistencies.
- Incorporate player auction/salary data to study the relationship between a player’s price and their on-field performance.
- Build a predictive model to estimate win probability based on toss outcome, venue, and historical head-to-head results.
- Automate the data refresh so the dashboard updates directly as new match data becomes available.

14. References

1. Dataset: IPL Seasons 2008 to 2026 Dataset, Kaggle. <https://www.kaggle.com/datasets/slidescope/ipl-seasons-2008-to-2025-dataset>
2. Microsoft Power BI Documentation. <https://learn.microsoft.com/power-bi/>
3. *[Add any additional articles, books, or research papers you referred to, in APA or IEEE format.]*

15. Appendix

A. DAX Measures Used

Measure	DAX Expression
4s	<code>CALCULATE (COUNTROWS (ball_by_ball_data) , ball_by_ball_data[batter_runs]=4)</code>
6s	<code>CALCULATE (COUNTROWS (ball_by_ball_data) , ball_by_ball_data[batter_runs]=6)</code>
Num of Balls	<code>COUNTROWS (ball_by_ball_data)</code>
Num of Overs	<code>COUNTROWS (ball_by_ball_data) / 6</code>

B. Key Calculated Columns

Column	DAX Expression
LosingTeam	<code>IF (ipl_matches_data[match_winner] = ipl_matches_data[team1] , ipl_matches_data[team2] , ipl_matches_data[team1])</code>
Losing TeamName	<code>RELATED (teams_data[team_name])</code>
IsFinal	<code>IF (CONTAINS (IPL_Finals, IPL_Finals[id] , ipl_matches_data[match_id]) , "Final" , "Non-Final")</code>

C. Dashboard Layout Summary

Page / Element	Details
Page 1: Dashboard	Title, 4 KPI cards, 1 bar chart, 1 table, 2 clustered column charts, team logo images
Page 2: Dashboard Part - 2	Title, 6 KPI cards, 2 charts, 6 slicers (Season, Venue, City, Result, Batter, Team)

[Insert additional dashboard screenshots, any SQL/Python code if used, and extra supporting charts here.]

16. Acknowledgement

I would like to express my sincere gratitude to my teachers, Mr. Sumit Kumar Mitthu and Mr. Harpreet Singh Bedi, for their valuable guidance, encouragement, and continuous support throughout this project, undertaken as part of my summer internship. Their insights and feedback were instrumental in helping me understand the practical aspects of data analytics and dashboard design, and in shaping this project from data collection through to the final dashboard and report.

I am also thankful for the opportunity to work on this project, which allowed me to strengthen my skills in Power BI, data modelling, and data-driven storytelling, and to apply classroom learning to a real, publicly available dataset.

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